NUS-ISS

MASTER OF TECHNOLOGY IN SOFTWARE ENGINEERING

Graduate Certificate Examination
Sample Examination Paper

Subject: *Architecting Scalable Systems*

SECTION A

Question 1

(Total: 25 marks)

Refer to the case study in **Appendix A** and answer the following questions:

a. One of the measures of success of a platform is how it grows its ecosystem by growing its reach with new types of interactions and number of users.

i. During the testing of Phase 1 API request-response type of interactions for the '***Verify Identity***' use case, the response time is expected to be less than 3 seconds for 90% of the requests for 100,000 voters.
Additional parameters of performance testing:
1. **API calls time frame: 30 seconds**
2. **Maximum response time: < 5 seconds**

A few testers had used some performance testing tools, e.g., Postman, to conduct the above test. It was however found that test results were varied and indeterministic, hence not a good proxy for the above performance criteria. Provide **one possible reason** for this observation. Propose **one more reliable way** to perform the test.

(2 Marks)

ii. The following interaction '***Update Current Simplified/Detailed Results***' use case requires update of current simplified results should be completed within 1 minute on average while update of current detailed results should be completed within 3 minutes on average for the 100,000 concurrent voters planned for Phases 1 and 2. Furthermore, the results produced by the use case have to be deterministic and repeatable.

An aspiring Solution Architect had chosen to make use the message queue mechanism for triggering the above use case whenever a vote is concluded in the *Conduct* Vote use case. As the Lead Solution Architect, you strongly recommend the use of the event streaming mechanism instead. Provide **two reasons** to justify your recommendation.

(2 Marks)

b. With reference to the domain model found in Section 7 of the case study, analyze the following issues. For **each** issue, identify the **most likely cause**, propose an effective **countermeasure** (by revising the domain model) and suggest the major **drawback(s)** of the countermeasure, if any.

i. The NotificationManager service is responsible for defining all kinds of notification used by the platform. The various kinds of notification are rather specific to the responsibilities of the other services of the platform. As the Lead Solution Architect, you are of the opinion that the current domain design does not facilitate the potential reuse of the NotificationManager service in other platforms. Significant cost saving could be realized if the NotificationManager service can be easily reused in other platforms without drastic adaptation.

(5 Marks)

ii. As the Lead Solution Architect, you are pessimistic of the ability of the implemented platform in allowing the Observers in promptly viewing the updated CurrentSimplifiedResults of the Poll he/she is monitoring.

(4 Marks)

c. An API framework provides the tools to build, expose and consume web APIs to enable the integration of different functionalities in this ***Vote Anywhere*** platform. It is important to be compliant with the Richardson Maturity Model when designing REST APIs.

During the first phase, development of the following APIs for the platform were selected; and your team designed the APIs for these two use cases: (i) *Invite Next Voter* and (ii) *Start Vote*.

The following interfaces illustrate the current design of the two APIs based on the RESTful approach (c.f., Richardson Maturity Model - Level 2).

URI	Service	Inputs	Outputs	Explanation	HTTP Action
/v1.1/votes/{voter-id}/invite	ProctoringManager	Voter identifier	List of voters in the polling room	Invite the next voter to the polling room for verification	GET
/v1.1/votes/{voter-id}/start	ProctoringManager	Voter Identifier	List of candidates of the poll, list of voters in the polling room	Put a voter to voting mode	GET

It is essential for API design to conform to the standard model for ease of understanding, usage and audit purposes.

With reference to the interfaces provided above, answer the following questions:

1. Do the above API interfaces comply with RMM Level 2? Provide the **likely reason(s)**.
2. Propose **a better versioning solution/strategy** to ensure API consumers are less implicated by API upgrades.
3. There are redundant and/or missing inputs, outputs and possibly wrong HTTP verbs in the API interfaces. Reconstruct the above table to depict the corrected interfaces.

(6 Marks)

d. A team member has suggested that the usability of the system, particularly in enabling proctors to manage voting sessions effectively, is the **most crucial non-functional requirement**, especially considering the high volume of voters. Do you **agree with** this proposal, and if so, why? If not, what are your reasons for disagreement?

(2 Marks)

e. A team member has suggested monitoring the **daily number of voters processed by the system** as an application metric. The proposal argues that tracking this metric will

help assess system **scalability** and **usability**, especially when the number is high. Do you **agree with** this proposal, and if so, why? If not, what are your reasons for disagreement?

(4 Marks)

END OF SECTION A

SECTION B

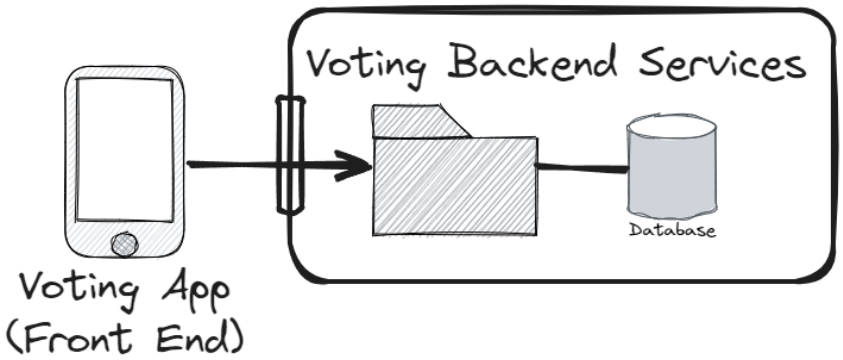
Question 2

(Total: 13 marks)

With reference to the case study in **Appendix A**, answer the following questions based on the following context:

The cloud native engineering team decided to develop a smart frontend *(Hinted but not limited to Phase 2 in the case study)* and these improvements will be part of *Vote Anywhere* platform products. As a cloud native architect, you are responsible for a few architectural and design decisions involved in both Phases 1 and 2. Answer the following questions with the above-mentioned backdrop:

a. Initially, your team created an initial architecture for its minimal viable product prototype as below.



Improve the above architecture by transforming the monolithic architecture into a hybrid reference architecture hosting relevant backend services needed by the voting platform *(hint: you can consider using microservices, scheduled tasks, event/stream-driven services and serverless).*

i. Capture the essential components and their interactions from a cloud perspective in a reference **architectural diagram**. What are the additional components or managed services that are essential for such a multi-cloud environment?

ii. In addition to external communication via API Gateway, how will the internal services communicate? Label the communication components in the diagram and explain the communication patterns preferred.

iii. Identify any **one** out of the designed services that may need to remember state. Which state-management mechanism would suit the service you have identified?

iv. Describe an overall deployment plan in terms of Kubernates cluster either using the k8s deployment diagram or as a paragraph.

(10 Marks)

b. Observe the logical architecture diagram provided in Section 6.2 of the case study. As mentioned, the voters front end app needs native local storage back up facility. In the near future, the platform plans to expand in other countries with provision for physical voting booths involving additional IoT sensors and actuators. In such context, would

you develop the proposed front-end using a ***native client or as a progressive web app***? What are the design implications on the backend services.

(3 Marks)

END OF SECTION B

SECTION C

Question 3

(Total: 12 marks)

Refer to the initial architecture specified in Section 6, the domain model in Section 7 and the logical packaging cum deployment model for Phase 1 in Section 8 of the case study in **Appendix A** and answer the following questions:

c. With reference to the domain model in Section 7 and the **incomplete** and **potentially erroneous** logical packaging cum deployment model in Figure 6 in Section 8, as the Lead Solution Architect, evaluate the logical architecture and identify **two undesirable architectural decisions** and propose the **corrective actions**. **Justify** your answer by explaining how the undesirable architectural decisions lead to the proposed corrective actions.
Hint: You may draw just the necessary parts of the diagram to illustrate your answers.

(6 Marks)

d. During performance testing, it was discovered that the system cannot handle the required 100,000 concurrent voters casting their votes within a 5-minute window due to a bottleneck in database write throughput. The current database setup includes one primary replica and one read replica. Please propose a comprehensive plan to scale up the database write throughput, including any necessary changes or impacts on software and infrastructure components. Additionally, identify at least two potential side effects of implementing this new design that the team should monitor closely.

(6 Marks)

END OF SECTION C